approximation is valid so the USAXS intensity shows an inflection proportional to $-\exp(q^2R_g^2/3)$ where R_g is the radius of gyration. At higher q values, the USAXS intensity follows a power law slope with an exponent of −3, as expected from Porod's law from slit-smeared USAXS. Our USAXS could not resolve the Guinier inflection from the large pores made by polymer microbeads, so we used a combination of a powder law slope and a log-normal size distribution of spherical pores to model the USAXS intensity for non-linear least squares fitting.

The scaffolds and impregnated anodes were imaged by FEG-SEM (LEO Gemini 1525).

Electrochemical measurements were carried out with an Autolab PGSTAT. Symmetrical cell tests were carried out under humidified hydrogen in nitrogen, while fuel cell tests were carried out under humidified hydrogen in nitrogen and humidified syngas in nitrogen (15% H_2 , 25% CO, 60% N_2). The exact syngas composition with a high content of nitrogen was chosen so that some of the nitrogen can be replaced with other components (*e.g.* CO_2 , larger quantities of steam, tars or H_2S) in later studies. For the syngas tests the cells were put under load in 50% H_2 in N_2 before switching to syngas and carrying out the measurements. This avoids the carbon deposition caused by direct exposure to syngas at OCV. In all cases the same procedure for reducing the anode was employed; the cells were heated under dry nitrogen to 765 °C at 5 °C min^{-1} before switching to humidified 10% hydrogen

in nitrogen for one hour before cooling to the operating temperature. In all cases in this report, the humidification was carried out at 20 °C, giving a humidity of 2.3%, while the gas compositions given refer to the gas compositions prior to humidification. [...]

练习5的答案

Nanoparticle scaffolds for syngas-fed solid oxide fuel cells[n]

Experimental

Fig. 1 **summarises** schematically the formulation and fabrication procedures **used in the paper**. Inks **were formulated from** a mixture of commercial gadolinia-doped ceria ($Ce_{0.9}Gd_{0.1}O_{1.97}$,CGO,particle size = 0.5 μm) **from** Praxair, nanoparticle CGO ($Ce_{0.9}Gd_{0.1}O_{1.97}$,particle size = 5 nm **from** TEM) **produced by** continuous hydrothermal flow synthesis (CHFS), polymer microbeads (6 μm, Microbeads AS, Spheromers CA6), terpineol as a solvent, Hypermer KD15 as a dispersant and an ethylcellulose binder (Hercules EC N-7). The CGO nanopowders **were produced by** continuous hydrothermal flow synthesis **using** a pilot plant **at** University College London **as described elsewhere.**[17] CHFS **is an emerging and scalable method of** preparing nanoparticles and **has previously been used** to make samaria-doped ceria,[3]

ceria-zirconia solid solutions,[18,19] an entire phase diagram of 66 ceria–zirconia–yttria samples[20] and a range of other metal oxides[21,22] and phosphates[23] **with** typically sub 50 nm diameter. **In** CHFS, an aqueous mixture of cerium ammonium nitrate and gadolinium nitrate solution **was continuously mixed with** a stream of superheated water **at** 450 °C and 22.1 MPa **in** a patented confined jet mixer.[24] Particles **were then cooled** (***via*** a pipe cooler) and **then collected as** an aqueous slurry **at** the exit **of** the back-pressure regulator. The particles **were recovered by** centrifugation and washing **(three times) in** an equal volume of water **until** the final solid sludge **was obtained. The** nanopowder **was recovered** by freeze drying.

For preparation of scaffolds, the solvent (0.653 g), dispersant (0.265 g) and binder (0.040 g) **were mixed** together **until** the binder had **fully** dissolved. **Next, with** stirring **in** an ultrasonic bath, **each** solid component **was added** sequentially **in** two or three aliquots, **starting with** the dried CGO nanoparticles (0.751 g), **followed by** the commercial CGO (1.047 g), and **finally** the polymer microbeads (0.153 g). The ultrasonication and stirring **is effective in** suspending the nanoparticles. This **gave** an ink **with** *ca.* 26 vol% solids content, **which was an acceptable level for** a screen printing ink.[11] **In** formulating the ink, two conditions **were sought**: high total solid content and high nanoparticle content. The order of addition **was important to ensure that** the nanoparticles were **fully** incorporated and homogenised, and the large differences in particle size **between** the different

components **helped to maintain** an **appropriate** viscosity **at** high solid loadings.

The inks **were screen printed onto** 200 μm thick yttria-stabilised zirconia (YSZ) electrolytes. In the case of fuel cells, the procedure **differed** slightly **between** cells **with** different sintering temperatures **for** the anode, as shown schematically in Fig. 1. **In** the cells **with** scaffolds sintered **at** 1300 °C, one side **was printed with** the scaffold ink, while the other **was printed with** a thin layer of a CGO ink (**this is to prevent** reactions **between** the cathode material and the YSZ electrolyte). The printed pellets **were then heat-treated at** 1300 °C **for** one hour, **with** a 1 °C min^{-1} ramp rate during heating and cooling. The LSCF-CGO cathode (ink **purchased from** Fuel Cell Materials, Columbus, OH, USA) **was then screenprinted**, and the cell **heat-treated at** 1100 °C **for** three hours **with** a 5 °C min^{-1} ramp rate during heating and cooling. The process **for** the cells **with** scaffolds heat-treated **at** 1000 °C **was identical except** the order of printing and sintering **was** a protective CGO layer **first, followed by** cathode and **finally** by the CGO scaffold. Cells **with** one, two or three layers **of** CGO scaffold **were produced**. The scaffold **was dried in** an oven **at** ~80 °C **in between each** printing cycle.

Next, the CGO scaffolds **were impregnated with** a nickel salt, again shown schematically in Fig. 1. The impregnation solution **was prepared from** 0.125 g **of** nickel nitrate hexahydrate **in** 200 μL **of** absolute ethanol. The solution **was dropped onto** the scaffold **using** a 1–20 μL

pipette, **with** the volumes used **determined** empirically **by** visually observing whether the drop spread **beyond the edge of** the scaffold. If it did, then the volume **of** the following impregnation **was reduced**. The first volume used **was** determined by dropping pure ethanol **onto** the scaffold. **In between each** impregnation, the pellets **were placed into** a furnace **directly at** 500 °C **for** 5 min. This cycle **was then repeated** ten times. **After** the final impregnation, the pellets **were placed into** this furnace, and the temperature **was increased to** 800 °C **at** 5 °C min^{-1} and **held for** eight hours **before** cooling **at** 5 °C min^{-1}. The NiO loading **was determined by** weighing.

Four symmetrical cells **were produced**: a single layer cell heat-treated **at** 1300 °C; a single layer cell heat-treated **at** 1000 °C; a double layer cell heat-treated **at** 1000 °C and a triple layer cell heat-treated **at** 1000 °C. In addition, four button cells **for** fuel cell testing **were produced from** single and double layer CGO scaffolds heat-treated **at** 1000 °C, and the same heat-treated **at** 1300 °C.

The scaffolds **were studied by** Ultra Small Angle X-ray Scattering (USAXS) **to obtain information about** the pore structure, while the *in situ* measurements **allowed us to** monitor the development **of** the microstructure **with** temperature.[25] **For** the USAXS measurements, single layered samples **were printed onto** MgO substrates (used instead of YSZ **as it is** less absorbing **of** X-rays) **for** analysis **by** USAXS, and **heat-treated at** either 1000 or 1300 °C. These

samples **were compared to** a scaffold **made from** a "control" ink **in** which the nanoparticle component **was replaced by** extra commercial CGO. One sample printed **on** MgO **was heat-treated** *in situ*. The USAXS data **were taken at** sector 15, Advanced Photon Source, Argonne National Laboratory. **Briefly**, small angle scattering intensities around the direct beam **was measured using** channel cut Si crystal monochrometers **in** a Bonse–Hart configuration. The X-ray energy **was** 16.8 keV and the beam size **was** 1.5 × 0.4 mm. The sample **was mounted in** a LinkAm heater **for** the *in situ* USAXS measurements. The USAXS intensity **is plotted** as a function **of** the scattering vector $q = 4\sin(\theta)/\lambda$ where λ is the X-ray wavelength and θ is the scattering angle. The USAXS spectra **can be viewed as** the autocorrelation **of** the electron density fluctuations **inside** the sample, resulting **from** contrast **between** the pores and the powders. The USAXS **is normally plotted in** a log–log plot, and **at** low q Guinier approximation **is** valid **so** the USAXS intensity **shows** an inflection proportional to $-\exp(q^2R_g^2/3)$ where R_g is the radius **of** gyration. **At** higher q values, the USAXS intensity **follows** a power law slope **with** an exponent **of** −3, as expected **from** Porod's law **from** slit-smeared USAXS. Our USAXS **could not resolve** the Guinier inflection **from** the large pores **made by** polymer microbeads, **so we used** a combination **of** a powder law slope and a log-normal size distribution **of** spherical pores to model the USAXS intensity **for** non-linear least squares fitting.

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撰写成功的方法部分的提示

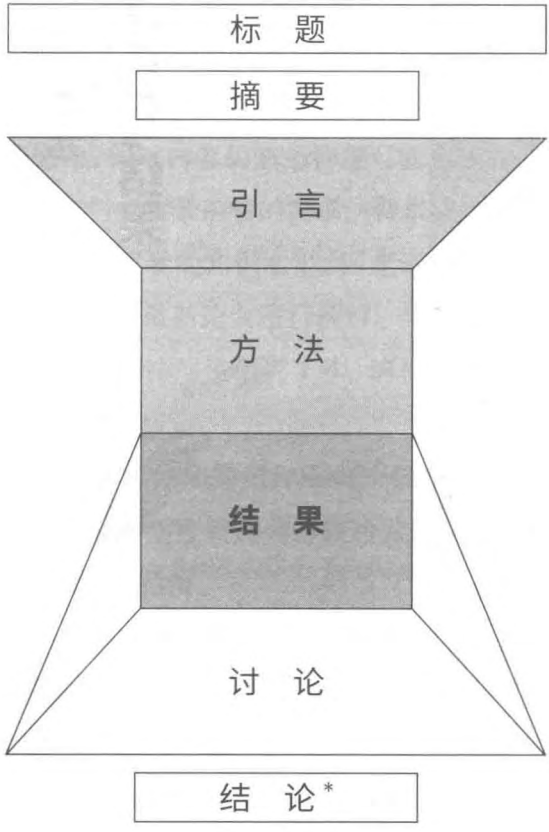
- 检查目标文章中方法部分的平均长度。这将帮助你确定适合描述该类型的实验或模拟的详细程度。

- 检查方法部分通常在目标期刊中出现的位置。它是在离线补充材料中还是在印刷版文章中?
- 在开始创建句子之前,请计划整个方法部分。
- 不要将你的实验记录与方法部分混淆。前者是为你而写的,后者是为读者而写的。
- 请记住,你对内容的熟悉程度可能意味着你不了解读者对入门或背景信息的需求。背景事实信息的数量取决于当前读者的知识水平,因此请与有关类似期刊中类似主题的最新论文进行核对。如果更多的读者可以通过互联网访问该研究工作,请相应地调整介绍性或背景信息的数量。
- 确保读者准确地知道你在每个阶段的所作所为的原因。这会产生连贯阅读的叙述方式,并支持你作为研究人员的诚信度。
- 现有/当前方法的参照范围以及描述这些方法的动词时态取决于当前读者的知识水平,因此,请与相似期刊中涉及相似主题的最新论文进行核对。
- 检查每个句子,以确保动词时态表达了正确的含义。
- 确保你将自己的研究成果与其他研究人员的研究成果和已有的知识区分开来。你的贡献对你来说可能很明显,但并不是每个读者都像你一样清楚明白。使用第2.5.1节中的无施动者被动语态表来检查你是否避免了任何潜在的歧义。

第三单元

怎么写结果





* 有些期刊在这一部分使用结论一词的单数形式，有些使用复数形式，但这似乎并不能反映所得结论的数量。

3.1　结果的结构和内容

下表列出了研究论文最后一部分的四个常见选项。

子标题选项1	子标题选项2	子标题选项3	子标题选项4
Results	Results	Results and Discussion	Results
Discussion	Discussion		∅
Conclusion	∅	Conclusion	Conclusion

你的目标文章会帮助你在这些选项之间做出选择，但是如果目标期刊的论文结构很灵活，那么你的研究工作可以指导你的决定。如果你的初步结果会影响以后的结果，那么可以倾向于采用结果和讨论（Results and Discussion）的组合，先介绍和讨论初步结果，然后再介绍和讨论后面的结果。类似地，如果你的结果对应于多个假设/预测，那么对读者来说看到每个假设对应的结果和讨论以及相应的子标题是更容易的。如果你打算对结果进行整体讨论，你可以将所有结果一起呈现在结果（Results）部分，然后转到讨论（Discussion）部分。最终目的是尽可能地使这个部分更容易让读者理解，以证明结果与研究目的之间的关系，并为结论（Conclusion）开辟出一条明确的路径。

在本书中，结果部分将与讨论部分分开处理，讨论部分将与结论部分分开处理。这是为了确保你有机会训练自己识别并创建每个部分特有的组件，无论是将它们分开还是组合起来。

请注意在某些情况下，根据期刊、主题或研究类型的不同，结果部分可能被称为数据分析（Data Analysis）或案例研究（Case Study）。与往常一样，用你的目标研究文章作为指导是行动的黄金法则。

3.2 创建模板

有关作者在结果部分应该和不应该包含的内容的建议通常与成功发表的内容相冲突。例如，你可能被建议避免在结果部分中对数据进行任何诠释。你可能被建议仅使用过去时，或避免使用重复的段落结构。你可能会被告知结果部分应仅写结果是什么，而讨论部分应仅写结果表达了什么意思。但是，当我们检查和描述成功发表的研究论文时，发现作者并不总是（甚至经常不）遵循这样的建议。

一个基于当前已经发表的研究论文的逆向工程的叙述方法，将生成更可靠的关于结果部分的描述，而不是那些告诉你应该或不应该做的规定。原因之一是结果部分的结构和内容正在发生变化，反映出读者的阅读方式，尤其是他们使用互联网作为资源的方式。当前在补充材料中放入方法部分的趋势也正在影响结果部分的内容，因为这意味着结果部分将需要包括有关方法的信息，以使结果更易于理解。因此，规定性建议可能很快就会过时。相比之下，逆向工程的叙述方法使你可以随着时间的推移轻松地更新和调整你的写作模板。

为自己开发逆向工程的方法意味着你可以将其应用于任何类型的文本，无论是现在还是将来的。在本单元中，我们鼓励你自己去控制分析和建立模板的步骤，并使用你的目标研究论文作为主要资源，建立自己的结果部分的模板。目的是建立**你所在领域的当前研究论文的结果部分的模板，并确保你能发展出更新和自信地适应它的能力**。

3.2.1　练习1：使用目标论文建立模板

在大多数情况下，研究工作的结果可以以图表、方程、照片或其他图像（在此均称为图形）的形式呈现。那么为什么研究人员还要这么麻烦去写一个结果部分呢？为什么不简单地提供具有清晰标题的精美图形，以及每个标题下方的注释呢？目前，几乎所有STEMM研究文章都包含一个正文的结果部分，因此很明显，结果部分的一些功能，如解释结果，没有叙述文本是无法实现的。

考虑到这一点，首先在至少**两篇**目标文章的结果部分对每个句子的功能进行简要描述，以生成广泛的潜在组件列表。然后在下面简化的结果部分对每个句子的功能进行简短描述，并将你的列表与此描述进行比较。你列表中的某些组件将类似于第3.2.3节的句子描述，其他组件将视你正在进行的研究、你所在领域的当前惯例或撰写的文章类型而定。合并这两个列表，为你自己当前研究领域的研究论文的结果部分生成一组可靠的

模型组件，你还可以在研究发展时进行调整或更新。

On-bicycle exposure to particulate pollution: using a portable device for accurate measurement

Results

1 Data obtained in previous cycling safety studies[1,5,7] used fixed monitors positioned at high-emission locations to measure weekday concentrations of black carbon (BC) and nitrogen dioxide (NO_2). ***2*** Those studies found no significant difference in BC and NO_2 emissions between cycle lanes shared with buses and cycle lanes separated from vehicle traffic. ***3*** In our study, BC and NO_2 concentrations were measured using a PEMS-43, a portable emissions measurement system developed by Noxious *et al.*[7]. ***4*** The PEMS-43 was fixed to a bicycle and recorded the emissions hourly along each route type.

5 Figure 1 shows the BC and NO_2 concentrations measured hourly on each route type from 6.00 a.m. to 10.00 a.m. ***6*** In line with results from previous studies, the BC and NO_2 concentrations were broadly similar for both route types over this 4-hour period. *7* However, the data obtained by PEMS-43 for journeys later in the day were significantly different. ***8*** We recorded a striking reduction in the NO_2 concentrations along bike paths separated from vehicle traffic, beginning shortly

after 10.00 a.m. (Figure 2). ***9*** As can be seen, by 2.00 p.m. the NO_2 levels had fallen by as much as 33%, and this low level was maintained even during the 5.00 p.m. to 7.00 p.m. peak traffic density period. ***10*** These results suggest that NO_2 levels along bike paths separated from vehicle traffic may not match those of BC at all times throughout the day.

11 The presence of vegetation is known to slightly alter BC and NO_2 concentrations at different times of day[11,18], and the effect of this along the two routes was not investigated. ***12*** Nevertheless, the data obtained here suggest that using a portable emissions measurement system to measure BC and NO_2 exposures hourly may provide more accurate information for traffic management strategies than traditional on-site measurement.

3.2.2　答案

句1和句2："Data obtained in previous cycling safety studies[1,5,7] used fixed monitors positioned at high-emission locations to measure weekday concentrations of black carbon(BC) and nitrogen dioxide(NO_2). Those studies found no significant difference in BC and NO_2 emissions between cycle lanes shared with buses and cycle lanes separated from vehicle traffic." **作者重复和回顾该文章之前提及的研究方法和发现。**

句3和句4："In my studies, BC and NO_2 concentrates were

measured using PEMS-43, a portable emissions measurement system developed by Noxious *et al.*[7] The PEMS-43 was fixed to a bicycle and recorded the emissions hourly along each route type."

作者简要总结了本研究中使用的方法。

▶ 为什么要从回顾文章之前提到的研究或总结我使用的方法开始?

提供与之前部分平滑过渡的材料或为进入新的部分做准备,这对读者是友好的。回顾或重温以前章节的关键部分,以确保所有读者以相似的基础开始结果部分。例如,有些读者可能回想不起来引言中提到的领域研究空白/问题或关键的研究工作,而这两者都是研究结果的重要背景。

在结果的开始回顾一下前面的部分,这有助于那些快速浏览信息的读者(例如直接从标题/摘要转到结果,并绕过引言和/或方法)。如果你查看目标文章中的结果部分,则可能会对本节中包含的方法所提供的大量信息感到惊讶。当方法部分很长或包含在补充材料中时,回顾和重温方法的关键元素是很常见的做法,这也确保了读者充分了解该方法以理解结果。你当然是对方法了如指掌的,毕竟这是你自己的研究,但你的读者并不像你一般熟悉。他们可能没有读过方法部分,或者在阅读结果时没有记住关键内容。

为什么不能以按顺序排列研究结果开始?

一篇论文的目的并不是按照事件发生的顺序来讲述故事，而是要对你的领域做出有效的贡献，并高效能地传达这一贡献。读者在阅读时会在结果中寻找结果的趋势和模式。如果你仅仅按照结果发生的顺序列出，那么读者就需要不断地调整该趋势或模式来确定解读结果的方向。你知道这些结果的去向，但读者却不知道。缺乏有效的叙述指导，结果的方向可能是不清楚的。此外，不同的读者可能会在数据中看到不同的模式，其中某些模式可能会使它们远离（而不是朝向）你自己的解释。身为作者的你有责任在一个叙述中组织和呈现结果，从而以最佳的方式构造它们，并自然而然地得出你想得出的解释和结论。

那么什么是好的开始方法呢?

和往常一样，查看目标期刊中成功的作者是如何开始的。一些作者以关于结果的整体模式或趋势的简短综述开始（In most cases/In general/Overall），并可能把读者引导到查看一个汇总性的图或表格。以概述作为开始，这可以稳定你要在其中展示结果的框架，并确保读者在看到单个“砖块”之前就已经看到“墙”。如果要将结果细分为多个结果的集合，你可能需要提供一个链接说明以引入每个结果。

有些作者从重新陈述研究目的开始。在讨论或结论中，你将讨论你的研究在多大程度上实现了引言中设定的目标。在结

果部分的开始，重启目标或问题，可以为读者设置评估你的结果的背景。当你决定要在结果中包括什么以及以什么顺序排列时，清楚地了解研究的目标对你也是有帮助的。你的结果应与你的目标明确相关，实际上，当你查看结果时，可能会决定回到引言，并根据结果重新定义或重述你的目标。

其他作者会首先回顾方法的关键方面。如上所述，当方法部分只能通过补充材料获得时，这种回顾尤其常见。它也可以让读者直接从标题转到结果，或从标题和摘要转到结果。

句5："Figure 1 shows the BC and NO_2 concentrations measured hourly on each route type from 6.00 a.m to 10.00 a.m." **作者邀请读者看一下图表的结果。**

▶ 为什么要邀请读者看图表?

读者在继续阅读时会看到它们，他们可能在开始阅读论文之前就已经看到过图形了。

当你阅读论文并且遇到句5之类的句子时，你会怎么做？在大多数情况下，你会停止阅读并查看图形，试图理解图形或诠释所看到的数据，然后你带着脑海中那个诠释回到文本并继续阅读。

将图形放置在特定的位置并叙述其框架显然很重要，要不然，它们将在结果部分的开始或结尾时以随机顺序出现，或者在整个结果中随机出现。为了最佳地放置图形，请在目标文章的结果部分中查看图形的顺序和位置，以及图形前后的叙述文

字。考虑作者为什么要按此顺序放置图形，以及你，作为读者，如何对这些图形的位置、顺序和叙述文本的框架做出反应。你可能会不耐烦地要呈现自己最重要、最激动人心的结果，但从以图形和叙述作为基础或引出更重要的内容开始可能会更有效，以便读者可以与你一起“发现”重要的内容。

有时，概述数据或调查结果的图形会显示在结果部分最开始的文本以前，尤其是在数据看起来非常清晰且无歧义的情况下。然而，如果图形、表格和照片中数据的直观表示不只有一种阐释，你最好在邀请读者浏览结果之前对数据进行评论。如果不这么做，读者和你对结果的阐释可能会有出入。

句6和句7：“In line with results from previous studies the BC and NO_2 concentrations were broadly similar for both route types over this 4-hour period. However, the data obtained by PEMS-43 for journeys later in the day were significantly different.” **作者使用主观的、评价性的语言将结果与其他研究的结果进行比较**（in line with、broadly similar、significantly different）。

我需要把自己的结果跟其他研究者的结果相比较吗？

在你开始进行原创研究时，你就越过了一条无形的线。到达这条线以前，你是在学习，在为老师或课程负责人而写作，这些人比你对主题的了解要多，并且他们知道期望在你的报告中看到什么。你的任务是向读者描述你的方法和结果，这些读者已经知道你会应用什么方法，会获得怎么样的结果，以及这些结果

和其他研究结果的比较。与之相反，研究的一个明确特征是它必须对该领域做出新的贡献，并且你必须明确地传达这一贡献。

你的研究和已有文献之间的关系是自始至终贯穿整篇研究论文的主线。在结果部分，将你的结果与文献中的结果相比较，呈现出你的结果如何引领或者改进这一研究领域的知识，使读者能够在研究地图上定位你的工作。这个研究地图在讨论中继续发展，为了有效地做到这一点，你首先需要把你的结果和已有的结果进行比较。

在结果部分中可以找到许多类型的比较。除了将结果与文献中的结果进行比较，你还可以将自己的结果与预测结果进行比较，或者将建模/模拟结果与实验结果进行比较。用于传达这些比较的相似性或差异性的语言可能对文本的成功至关重要，你可以在第3.4.2节：**将你的结果与其他结果/建模结果/预测结果比较**一表中找到大量选项。

为什么必须使用评价性的语言——为什么不能简单地描述图形的内容?

在结果部分要使用描述性的语言，而不是仅依赖图形，最重要的原因是结果不能自己开口阐述。你并不是简单地撰写结果，而是撰写**一个关于结果的指南，它应该有逻辑地和自然地引导你将得出的结论。**

你用来描述结果和对结果发表评论的语言会影响读者解读它们的方式。如图3.1中的两条曲线：

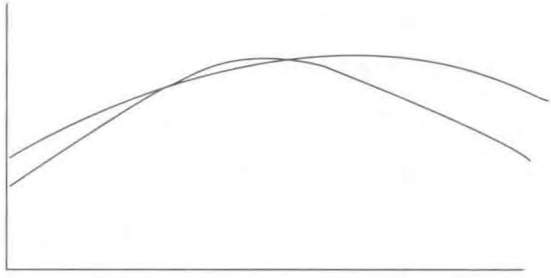


图 3.1

如果你写作“如图3.1所示，两条曲线非常相似”，读者将专注于两条曲线之间的相似之处。然而，如果你写作“如图3.1所示，两条曲线明显不同”，读者将专注于其不同之处。

你对结果的描述支撑了研究目标、结果本身和你的结论之间的一致性。如果你不对结果发表评论，使得读者可以共享你对结果的理解，那么他们的理解可能会与你的不同，而你最终得出的结论对读者而言会变得既不合逻辑，也不自然。事实上，它们甚至会是令人吃惊或者不合适的。例如，如果你只写：如图1所示，23%的案例产生了效果（“As can be seen in Fig. 1, the effect occurred in 23%.”）。你可能会基于这样一个看法而得出你的结论，即23%的案例是比较高的比例。但是，如果没有任何评价性的语言来“固定”数字的价值，读者可能会独立地认为23%的比例是比较低的。这将对你论文的其余部分，尤其是结论，造成破坏性的影响。

尽管结果有时会被客观地描述出来（例如用数字或使用非

评价性的语言），但如果你仅仅描述图形中有什么，却没有给读者已经看到的内容添加任何东西，又何必要多此一举呢？作为科研队伍的合格成员，你有权也应当提供你对所得结果的理解。读者未必同意你的意见，但是他们需要知道结果的去向，否则你的结果和你从结果中得出的结论之间的差距将会变大而无法协调。因此，结果通常包括一些评价性的语言。

评价性的语言可能与相关的数字/数量一起出现（例如 only 23ml），或者可以代替数字/数量（例如in many cases）。你选择的语言将引导读者了解你自己对数据的诠释，因此，如果你要寻找的效果在23%的情况下会发生，你可以选择将其传达为强有力的结果（例如 in as many as 23% of cases）或者是比较弱的结果（例如 in only 23% cases）。有趣的是，研究人员常常否认他们自己领域的论文作者是这样做的——直到目标文章中的评价性语言被明确指出。你可以在第3.4.2节：**特定的/关键的结果**中找到通常用于描述和评论结果的评价性语言列表。查看列表后，请为目标文章中的示例加下划线或做出标记。这将给你一个论文中包含了什么样的、什么时候、怎么样以及多少评价性语言的明确指示。

请注意，在句7中，除了诸如 broadly similar 这样的评价性语言，作者还在句子的开头添加了明确的功能性信号（However,），以确保读者为看到差异做好了准备。这是一个简短的句子，因此信号放置在哪里都没有关系。在长句子中，最好将信号放置在句子的开始，因为以信号结尾的长句子会要求

读者“绕回”整个句子才能理解。比较下面的两个句子：

In such cases, the patient may require a higher dose of intravenous antibiotics, an intravenous load of phenytoin, or even a brief procedure under intravenous conscious sedation, unfortunately.

Unfortunately, in such cases, the patient may require a higher dose of intravenous antibiotics, an intravenous load of phenytoin, or even a brief procedure under intravenous conscious sedation.

句8：“We recorded a striking reduction in the NO_2 concentrations along bike paths separated from vehicle traffic, beginning shortly after 10.00 a.m. (Figure 2).” **作者把读者的注意力指向具体的结果上，并使用强有力的评价性语言**（striking）**对其进行描述**。

▶ 像“striking”这样的语言是不是看起来太不正式了？

显然不是。即使结果很令人兴奋，科研作者通常也不会用感叹号，而是用诸如惊人的（striking）之类的语言来表达那种“哇”的惊喜感觉。在讨论的语言部分中［第4.4.2节：（ii）“！”的代用词（“非常快乐的”词汇）］你会找到这些替代感叹号“!”的词汇列表。

句9：“As can be seen, by 2.00 p.m. the NO_2 levels had fallen by as much as 33%, and this low level was maintained even during the 5.00 p.m. to 7.00 p.m. peak traffic density period.” **作者选择一个具体的结果进行详细的描述，并对其进行评论**（as much as、even）。

我如何确定哪些结果需要详细描述？为什么不全部详细描述呢？

请记住，你的工作是提供一个**结果的指南**，而不仅仅是结果的描述。如果你详细介绍所有结果，那么它们看来都同等重要。但是情况并非如此：你知道某些结果是关键的结果，其他结果不那么重要，即使在同一组关键的结果中，某些结果也比其他结果更重要。最后，你的句子只是白纸黑字——读者并不能听到你的声音，因此并不能听到你在强调某个特定的结果。你不能像我们已看到的这样用红色把它打印出来，甚至也不能用感叹号。你的责任是选择重要的结果，并更详细地关注它们，这样你的读者就可以分享你对重要结果的解读。你还可以通过以如下语言开头的句子来阐明这一点："It is important to note that.../Importantly,/It is significant that…"

有趣的是，你会发现最好的结果往往以一种特别的方式被描述出来，以至于给人们留下它们是典型的结果的印象——这一点请在你读过的论文里留意一下。作者可以对结果做一个综述，紧跟着是一个类似 for example（例如）和 best result（最佳的结果）这样的短语："The SFS results are generally in very good agreement with their FE counterparts; for example, at midspan the values are almost identical."

我应该诠释结果并对其发表评论吗？

到目前为止，很明显这个决定只能通过仔细分析类似的当

前论文并进行逆向工程来做出。然而，你会看到这些诠释在大多数结果部分中都很常见。对**如何**获得特定结果的解释可能会包含有关你所用方法的详细信息。对结果**为什么**发生的解释可能包括有关你研究的材料属性的信息。考虑结果的**诠释**和结果的**含义**之间的区别（数据的含义或暗示）是很有帮助的。在结果部分，**诠释**往往仅限于对数据的直接解释；含义通常在结果中简要提到（请参见句10），但对结果的含义或暗示的完整讨论通常要留在讨论部分。

句10："These results suggest that NO_2 levels along bike paths separated from vehicle traffic may not match those of BC at all times throughout the day." **作者提到结果的可能含义**。

句11："The presence of vegetation is known to slightly alter BC and NO_2 concentrations at different times of day [11,18], and the effect of this along the two routes was not investigated." **作者提到了一个可能的局限性，将其对结果的潜在影响**（slightly）**减至最小**。

我需要在结果部分提到问题吗？这样难道不会让读者怀疑我的结果吗？

正如我们在方法部分讨论的（第2.2.2节句9），事实恰恰相反。除非你确定这些问题是不重要的和不明显的，否则不要忽视结果部分的问题。如果你的结果不完整或者某些结果并不"合适"，那么提及它的问题是很常见的。如果可以的话，将问题的重要性减到最小，并且暗示问题出现的可能原因或者解

决方法。如果你没有提到结果的问题或者缺口，说明你没有发现它们，这会向读者表明你并不完全了解自己的研究。相比之下，在研究中提及对问题的讨论则效果会恰恰相反：这显示了你对研究完全掌握，并且能够清晰地评价它。提及结果中的问题或缺口的另一个好处是，它为你提供了讨论/结论的关键要素：进一步研究的方向或建议——在这个例子中，建议进一步研究应该调查植被对排放物水平的可能影响（"a suggetion that future studies should investigate the possible effects of vegetation on emissions levels."）。

如前面所述，如果你想要等所有的结果都完美了再来提交要发表的论文，那么你可能永远都不能发表这篇论文。从实验室到期刊的速度至关重要。一旦你的结果值得传达，马上就可以写论文，撰写结果部分的时候需要提及并且承认你遇到的问题或困难。如果你在文章的讨论/结论部分要讨论将来研究的建议的时候，才第一次提到问题，这是不合适的。

那么我如何谈论结果部分的问题呢?

使用那些使问题最小化、暗示可能的原因和/或提出解决方案或者推进方式的语言。在上面的例子中，作者承认出现了一个问题并且使用一个词使其影响最小化（slightly）。你可以在第3.4.2节的问题和议题表格中找到提及不完美或有问题的结果的语言表达的例子。

句12："Nevertheless, the data obtained here suggest that using

a portable emissions measurement system to measure BC and NO_2 exposures hourly may provide more accurate information for traffic management strategies than traditional on-site measurement." **作者将读者的注意力从问题上转移开，转向研究结果的含义和应用方面的正面价值**(Nevertheless... may provide more accurate information)。

结果的含义和应用是否应该等到讨论部分之后才提及?

结果的含义是讨论部分的中心主题，然而几乎所有作者都会在结果部分中提及结果的含义。提出结果后，论文的焦点便开始展开，并从中心报告部分转向讨论和结论部分，因此在此刻经常看到像 suggest 或 indicate 这样的动词。这有助于读者查看论文的去向，并打开了通往讨论部分的路径。

提及结果的应用有助于读者将研究结果与引言或摘要中提及的目的或应用联系起来。作者可以使用与这些目标或应用相呼应的语言来创建这个连接。例如，如果引言中所述的目的是提高某些内容的准确性(improve the accuracy of something)，那么使用更准确(more accurate)的短语来描述结果将有助于读者清楚地看到这个连接。这些词汇和短语通常是非常积极的，有关示例，请参见第1.4.2节：4.当前的工作、第2.4.2节：证明选择的正确性、第3.4.2节：(iv)对结果的主观描述："高兴"字眼☺、第4.4.2节：(ii)"!"的代用词("非常快乐的"词汇)和第6.4节：成果/价值/贡献的"快乐词汇"列表。

3.2.3 结果模板

将下面的句子描述与你在3.2.1节中的列表进行比较。练习1：

句1和2，作者	重复/回顾文章前面提及的研究方法和发现
句3和4，作者	简要总结研究中使用的方法
句5，作者	邀请读者查看结果图形
句6和7，作者	使用主观的、评价性的语言将当前研究的结果与其他研究的结果进行比较
句8，作者	将读者的注意力吸引到具体的结果上，并以高度评价性的语言对其进行描述
句9，作者	选择一个特定的结果以更详细地呈现，并对其进行评论
句10，作者	提到结果的可能含义
句11，作者	提到可能的局限性，将其对结果的潜在影响最小化
句12，作者	将读者的注意力从问题上转移开，转向研究结果的含义和应用方面的正面价值

你可以简化这些步骤以创建基本的通用的结果部分模板（参见下面表格）。**将此模板与你在第3.2.1节中创建的组件列表合并在一起，以在你自己当前的研究领域的结果部分生成一组可靠的模板组件，你还可以随着研究的发展进行调整或更新。**例如，如果目标期刊中的方法部分仅可以通过补充材料链接获得，那么你可能需要在结果部分的开头对方法进行全面介绍。你的目标期刊可能会需要一个 Research in Context 部分以

展示与其他研究结果的比较。根据当前的结构模板提交你的研究，这表明你是能了解最新文献的研究人员。

通用结果模板	
1	回顾文献/目的/预测/假设/空白 回顾/总结方法 关于结果的宽泛性陈述 邀请观看图形和图形的内容
2	详细/关键结果±评价性语言/评论 与其他研究结果的比较/快乐的词汇 ☺ 与模型/模拟/预测结果的比较 通过已知事实/方法细节解释结果
3	结果的问题/缺陷±原因
4	结果的可能含义/快乐的词汇 ☺

注：
结果部分中的动词时态

- 描述在图像中可以看到的内容通常采用一般现在时态。这可能与描述“图像信息是如何获取的”的句子交替出现，而这些句子通常采用一般过去时态。
- 某些解释结果的背景事实的句子，例如所用材料的属性，可以用一般现在时态（例如，This occurred because material X is able to...）。此外，例如那些引用该方法的，可以用一般过去时态（例如，This occurred because we used a...）。
- 在第一单元中（**第1.5.1节动词时态的选择**）指出时态的选择可以表达作者对结果的永久价值或永久事实的信心。一般过去时态表示作者发现的事实；相比之下，一般现在时态反映了一种信念，即该发现足够可靠，可以构成一个永久性的事实。
- 结果部分可以自始至终使用一般现在时态，例如，描述模型是如何“运行”的。请查看你的目标文章。

结果部分中的小节/小标题

很多结果部分会被分为小节，对目标文章中的小节及其子标题进行仔细的逆向工程将为你提供有效的写作计划。首先观察一些简单的指标：目标文章的结果部分中有多少小节？这些小节有多长？每个小节中段落的功能是什么？子标题有多长？子标题包含语法句子吗？

在你完成几篇目标文章后，请考虑小节和子标题的功能。是什么在控制着各个小节的划分？每个小节的整体功能是什么？子标题和小节内容之间是什么关系？子标题是概述小节的内容吗？回答这些问题将使你能够计划你的写作，使你能够以体现你的领域中类似的结果部分的结构来进行写作。

3.3 验证和调整模板

3.3.1 模板演示

使用你的合并列表，在下面的结果部分中鉴定模板组件。表示模板组件的语言为**粗体**，在某些情况下动词为**粗体**，因为动词时态说明了模板组件。

Amazonian functional diversity from forest canopy chemical assembly[o]

Results

Regional Chemical Diversity. Canopy chemical traits **varied widely** among **the thousands of trees surveyed** along the Andes–Amazon elevation gradient (Table 1 and Table S3). Foliar N, P, and lignin **spanned** an order of magnitude in value, whereas Ca and phenols **varied** by two orders of magnitude. Community-scale variation in **many** chemical traits **tracked** changes in elevation (Fig. S1) and **at times, was closely related to** climate (Table S4). Intercomparison of elevational trends in canopy chemistry **was made possible by applying** a gradient normalization procedure to the data, which **shows** the percentage increase or decrease in a community's average trait value relative to the gradient mean **(SI Methods). By doing** this normalization, elevational trends among all forests **were found to differ from observed trends** among high-fertility sites alone, **revealing the central role of** soils in determining community level canopy chemistry in the region (Fig. 1). **Most notably**, foliar P and Ca concentrations on higher-fertility lowland sites **were** two times that measured on lower-fertility lowland sites, and soluble C concentrations **were elevated** in higher-fertility areas (Table 2). **In contrast**, total C, phenols, and lignin **were suppressed** in the higher-fertility sites.

We also discovered elevation-dependent tradeoffs in canopy foliar C allocation throughout the region. Up the elevation gradient, cellulose and lignin **decreased** 100% relative to their region-wide mean. Soluble C **increased by almost** 150% with elevation (Fig. 1), and **this change occurred in parallel to a nearly** 200% increase in LMA. Changes in C allocation **were tightly linked to** mean annual temperature and precipitation along the gradient (Table S4).

We found opposing patterns for P and Ca — two rock-derived nutrients **often thought to** limit growth in tropical forests (16). With increasing elevation, foliar P **increased 100%** above the gradient mean value (Fig. 1A), but this elevational pattern **disappeared** after the removal of the low-fertility sites from the analysis (Fig. 1B). **In contrast**, mean foliar Ca concentration **decreased by 100%** from the Amazonian lowlands to tree line in the Andes. Foliar N **declined only slightly** with elevation. Additional analyses **revealed** decreasing P and Ca on a leaf area basis, **despite the fact that** LMA **increased** with elevation (Fig. S2 and Table S5). Finally, foliar $\delta^{13}C$ **increased by about** 200% with elevation relative to its mean gradient value, and **this trend occurred** independent of site fertility (Fig. 1).

[...].Site characteristics **were a relatively small** contributor — less than 20% — to the explained variance in **most** canopy chemical traits (Fig. 2), **indicating that**, within any given community along the elevation gradient, phylogeny dominates over local differences in soils, microclimate,

and other factors. [...].Foliar $\delta^{13}C$ **displayed** the weakest phylogenetic partitioning. Canopy P and Ca patterns **were also dominated by** site conditions, especially soils; **this** soil fertility **effect is evidenced by the fact that** phylogeny **played a much stronger role** in determining foliar P and Ca when only considering high-fertility sites. Regressing the model components against elevation, **it is also clear that** the taxonomic partitioning of **most** canopy chemical traits **is** invariant with elevation (Table S6).

Inter- vs. Intraspecific Variation. Interspecific (between-species) variation in canopy chemical traits was consistently **two to three times greater** than intraspecific (within-species) variation, and intraspecific variation **was often very low** in canopy trees at all sites (Fig. 3 and Table S7). **Moreover, there were very few** elevation-dependent trends in either intra- and interspecific variation (Tables S8 and S9). Maximum intraspecific variation **was recorded** for Ca (24–29%), phenols (21–22%), and P (16–21%). $\delta^{13}C$, total C, and soluble C **showed extremely low** intra and interspecific variations of **less than** 10%.

3.3.2　练习2：识别模板组件

这是另外三个来自不同学科研究论文的结果部分。它们已被编辑以达到合适的长度，省略部分被标记为 [...]。用你的合并列表来确定模板组件。

1 Effect of muscle compensation on knee instability during ACL-deficient gait[1]

Results

In the absence of any muscular compensation, ATT increased throughout the stance phase of ACLd gait relative to that calculated for normal gait (Fig. 2). Peak ATT for the intact and ACLd knee occurred at contralateral toe off, which coincided with the occurrence of peak knee extensor moment (see Fig. 3B). Maximum allowable ATT was always greater than that calculated for normal gait, and at times was even greater than that estimated for ACLd gait (Fig. 2, compare lightly shaded region and solid line around 50% of gait cycle).

The model simulation results showed that it was not entirely possible to restore ATT in the ACLd knee to the amount calculated for normal gait merely by reducing the magnitude of quadriceps force (Fig.3A, gray solid line). There were periods near heel strike and in midstance when the lower limit of quadriceps force (zero force) was reached, and yet ATT in the ACLd knee was greater than that obtained for the intact knee (Fig. 2, black dashed line). The simulation results also showed that it was not entirely possible to restore ATT in the ACLd knee to the maximum allowable level by reducing quadriceps force alone (Fig. 3A, gray dashed line). In this case, there was a brief period during midstance when the lower limit of quadriceps force was reached, and yet ATT

was greater than that incurred during maximum isometric extension (Fig. 2, black dashed line exceeds the gray region during midstance).

Complete elimination of the knee extensor moment (a quadriceps avoidance pattern) was needed to restore ATT in the ACLd knee to the level calculated for normal gait (Fig. 3B, gray solid line). In contrast, some (positive) net extensor moment was predicted when ATT in the ACLd knee was restricted to the maximum allowable limit (Fig. 3B, gray dashed line).

The simulation results showed that it was possible to reduce ATT exactly to the level calculated for the intact knee merely by increasing the magnitude of hamstrings force (Fig. 4A). As expected, the increase in hamstrings force needed to bring ATT in the ACLd knee to the maximum allowable level was less than that needed to bring ATT to the intact level (Fig. 4A, compare gray dashed and solid lines). An increase in hamstrings force led to a decrease in the knee extensor moment, but this effect was noticeably less than that obtained when quadriceps force was reduced (compare gray lines in Fig. 3B and 4B). [...]

2 Identification and characterisation of the early differentiating cells in neural differentiation of human embryonic stem cells[c]

Results

Identification of a transient Tra-1-81(-)/SSEA4(+) cell population at early stage neural differentiation of hESCs

Similar to the cells in the ICM, undifferentiated hESCs express embryonic cell specific surface antigens, including stage-specific embryonic antigens (SSEA) 3 and 4, Tra-1-60 and Tra-1-81 [17], which were first identified in human embryonic carcinomas. However, unlike mESCs, they do not express SSEA1. Although the functional significance of these antigens is as yet unclear, they are routinely used as markers for hESCs. Therefore, we anticipated that studying the dynamic changes of these cell surface markers in neural differentiation could potentially enable us to capture and isolate the early differentiating populations which could be used for further analysis.

We focused our study on Tra-1-81, SSEA4 and SSEA1, given that SSEA3 expression is not necessarily required for a pluripotent state [18], and the high similarity between Tra-1-60 and Tra-1-81 [19]. Our results showed that, in self-renewal culture conditions, H1 hESCs express high levels of SSEA4 and TRA-1-81, but not SSEA1 (Figure 1A, top).

After differentiation to neural progenitors, expression of Tra-1-81 and SSEA4 were lost while expression of SSEA1 was increased (Figures 1A, bottom). However, it is noteworthy that the loss of Tra-1-81 preceded that of SSEA4. After 9 days in neural differentiation medium, Tra-1-81 expression was almost completely lost, whereas the majority of cells still expressed SSEA4 (Figure 1A, middle). The expression of SSEA4 persisted for several days until 2 weeks into the differentiation at which point the neuroepithelial cells started to emerge and SSEA1 expression was positive. A similar pattern was also observed in H7 hESCs, although Tra-1-81 expression was lost earlier than that in the H1 cells (7 days vs. 9 days) (Figure 1B). The existence of this Tra-1-81(2)/SSEA4(+) population of cells was also confirmed by immunostaining (Figure 1C). Further neural differentiation from this stage lead to the efficient generation of NPCs, which retained the expression pattern of Tra-1-81(2)/SSEA4(2)/SSEA1(+) and could be maintained for an extended time in culture when supplemented with bFGF/EGF (Figure 1D). The neural differentiation experiments were repeated several times in both H1 and H7 hESC lines and the sequential loss of the Tra-1-81 and SSEA4 antigens was reproducible, although the timing of the disappearance of each antigen varied slightly between experiments, depending on the initial seeding density.

To eliminate the possibility that the sequential loss of Tra-1-81 and SSEA4 is a culture-dependent phenomenon, hESCs were also differentiated using the double SMADs

inhibition protocol [20,21]. Differentiation with the dual SMAD inhibitors exhibited the same sequential loss of Tra-1-81 and SSEA4 in both H1 and H7 cells (Figure S1A). Furthermore, a previous report using stromal-feeder based neural differentiation protocol observed the same effect [19]. We therefore propose that this initial Tra-1-81(2) and SSEA4(+) population represent cells of early neural differentiation and consequently designated them TR2/S4+ cells.

TR2/S4+ cells exhibit an intermediate gene expression pattern between hESCs and neural progenitors

Different cell populations can be distinguished by morphological and developmental criteria, as well as by the temporal and spatial expression of marker genes. In order to characterize the transient TR2/S4+ cells, we isolated them by FACS or by magnetic-activated cell purification (Figure 2A). The Tra- 1-81(2) and SSEA4(+) identity of the purified cells was then confirmed by flow cytometry analysis. Gene expression was analysed by quantitative RT-PCR (qRT-PCR) and compared to that of undifferentiated TR+/S4+ hESCs and their TR2/S42 NPCs (Figure 2B). We found that the expression of pluripotent marker genes, Oct4, Nanog and Rex1, was clearly downregulated in the TR2/S4+ cells compared to hESCs but remained higher than that of NPCs. Conversely, the expression of neural progenitor marker genes, Pax6, nestin and Sox1, was upregulated in the TR2/S4+ cells but was considerably lower than in NPCs. However, Sox2,

another known pluripotent marker, was consistently expressed in all three cell types, with its highest expression level in the NPCs, implicating a critical role in both pluripotent and neural progenitor cells. In addition, the TR2/S4+ cells also expressed the early differentiation marker, FGF5, which was undetected in both hESCs and NPCs (Figure 2B). A similar gene expression pattern was also found in TR2/S4+ cells from neural differentiation with dual SMAD inhibitors. It was also found by RT-PCR that, compared to the undifferentiated hESCs and their NPCS derivatives, both the isolated TR2/S4+ cells and the unsorted day 9 differentiated H1 cells expressed much lower levels of leukaemia inhibitory factor receptor (LIFR) (Figure 2C).

Western blotting was performed to confirm the expression of Oct4 and Nanog proteins (Figure 3A). While undifferentiated hESCs exhibited high levels of Oct4 and Nanog proteins, they were undetectable in NPCs. However, in the TR2/S4+ cells, Oct4 and Nanog protein levels were lower than in hESCs but higher than in the NPCs. Since both RT-PCR and Western blotting methods quantify average levels of gene expression in a whole population, one cannot exclude the possibility that this TR2/S4+ population contains two groups of cells: one similar to the undifferentiated hESCs (high Oct4 expression but low/ no expression of neural markers), and the other similar to the NPCs (high levels of neural markers but no Oct4 expression). Therefore, to confirm that the expression of Oct4 and Pax6 in the TR2/S4+ cells was not due to the co-

existence of these two different populations, immunostaining and flow cytometry analysis were carried out. The results showed that Oct4 antibody staining was positive in 97% of hESCs, with almost 80% cells strongly positive (mean value =156.5). Similarly, 97% of TR2/S4+ cells were also stained positive for Oct4, but the majority of cells exhibited a lower level of Oct4 (mean value = 54.49, Figure 3B, left). No clear Oct4 staining was visible in NPCs (mean value =2.9). In addition, TR2/S4+ cells showed a clear upregulation of Pax6 signals (mean value = 12.4) compared to hESCs (mean value =3.14), which was not as strong as in the NPCs (mean value =21.1, Figure 3B, right). Co-staining of Oct4 and Pax6 in situ confirmed the flow cytometry analysis and clearly demonstrates the co-existence of Oct4 and Pax6 proteins in the same cell during early neural differentiation, which are notably expressed in a negatively correlated manner (Figure 3C).

To explore whether the various levels of Oct4 expression are regulated at the promoter, we examined the amount of DNA methylation in the Oct4 promoter region near the transcription start site. A gradual increase in DNA methylation was observed throughout the neural differentiation and was negatively correlated with Oct4 mRNA expression (Figure 3D). In hESCs, very low levels of methylation were detected (0.05%), opposed to the heavy methylation detected in NPCs (84.3%). Interestingly, an intermediate DNA methylation level of 48.1% was observed in the TR2/S4+ cells. Taken together, these results demonstrate that the TR2/S4+ cells exhibit a gene expression

profile which is distinct from both undifferentiated hESCs and their fully committed neural progenitors, and co-express both neural progenitor and undifferentiated pluripotent markers. In addition, Oct4 expression is progressively downregulated during neural differentiation, which is accompanied by the upregulation of neural markers.

TR2/S4+ cells generate both neural and non-neural lineage cell types

Under neural differentiation conditions [16], further culture of the TR2/S4+ cells lead to the efficient production of NPCs that express high levels of neural progenitor markers: nestin (97%), Sox1 (82%) and Pax6 (88%) (Figure 3 and 4) but lack expression of pluripotent genes (Oct4 and Nanog) or mesoderm and endoderm markers: GATA6, brachyury and a-fetoprotein (Figure 2) [22]. These neural progenitor cells were able to further differentiate into neurons and astrocytes as revealed by the positive staining of MAP2, TUJ1 and GFAP, respectively (Figure 4).

Since Oct4 expression persists in TR2/S4+ cells, albeit at a lower level, we considered whether these cells were able to generate non-neural lineages in alternative culture conditions. Purified TR2/S4+ cells were differentiated via cell aggregate formation, a method that can initiate spontaneous differentiation and is widely used to examine the differentiation potential of cells in vitro [23]. The TR2/S4+ cell aggregates were able to expand in culture and

displayed structures similar to embryoid bodies (EBs) after 7 days of differentiation in suspension (Figure 5A). These EBlike aggregates were then dissociated and plated onto adherent culture dishes for further differentiation. After another 7 days, cells with various morphologies were visible (Figure 5B) and were analysed for gene expression by qRT-PCR and immunocytochemistry. Real-time RT-PCR showed upregulation of markers of the three germ layers: neuroectoderm (Pax6, Sox1 and Sox2), mesoderm (goosecoid (GSC) and Meox1) and endoderm (albumin (ALB) and GATA6) (Figure 5C), while pluripotent markers, Oct4 and Nanog were downregulated. Immunocytochemistry further confirmed that the differentiated cells contained progeny from the three germ layers: AFP, HNF4a and GATA6 for endoderm, muscle actin for mesoderm, nestin and Pax6 for neuroectoderm (Figure 5D). These results demonstrate that TR2/S4+ cells are also capable of differentiating into cells of non-neural lineages.

TR2/S4+ cells cannot be maintained or reverted back to hESCs in self-renewal culture conditions

TR2/S4+ cells express a high level of FGF5, a marker of post-implantation primitive ectoderm of mouse embryos [24]. Given that mESC-derived primitive ectoderm-like cells are capable of reverting back to an ESC state when re-cultured in mESC self-renewal media [25], we examined whether theTR2/S4+ cells can be reverted back to their original hESC

state. Purified TR2/S4+ cells were re-plated into matrigel-coated plates and cultured in hESC self-renewal conditions (Figure 6A). To verify that any resulting changes were not a technical artifact, we purified and cultured TR+/S4+ hESCs as controls (Figure 6B). Both cell populations showed similar attachment efficiencies (Figure 6Aa&6Ba) and colonies were clearly visible 3 days after plating. By day 7, TR2/S4+ colonies exhibited structures and morphologies similar to that of hESCs but the expression of Tra-1-81 and SSEA4 could neither be reactivated nor maintained (Figure 6Ab). In contrast, control hESCs (TR+/ S4+) expressed high levels of both these markers (Figure 6Bb). This difference became even more apparent after 14 days of culture (Figure 6Ac & 6Bc). Consistent with the changes in cell surface markers, evident differentiation also started to emerge in the TR2/S4+ cells. RT-PCR analysis confirmed that expression of both Oct4 and Nanog was downregulated in TR2/S4+ cells as the culture proceeded. Mesoderm and endoderm markers were up-regulated while the expression of neural markers, Pax6 and Sox1, was reduced, although the levels remained higher than in hESCs (Figure 6C). These results are consistent with the data from EB cultures, which indicate that the cell fate of TR2/S4+ cells are affected by their culture conditions. In CM cultures, there is limited inhibition of BMP signaling in cells, at least not to the level induced by the addition of noggin [16], which may account for the observed nonneural lineage differentiation. These results further confirm that TR2/S4+

cells are distinct from undifferentiated hESCs and cannot be reverted back to their original hESC state by reintroduction into hESC self-renewal conditions. Furthermore, these cells cannot be maintained as TR2/S4+ cells under these conditions.

Comparison of TR2/S4+ cells with cells of early embryoid bodies and definitive endoderm differentiation

Since TR2/S4+ cells are able to generate cells in addition to the neural lineage, it raises the question of whether these cells are equivalent to those of early EBs. To answer this question, we compared our global gene expression data of early neural differentiation (N1), which resembled the TR2/S4+ cells (over 85% cells are TR2/S4+) [22], with the published microarray data from 16-day EBs of the same hESC line, H1 [26]. Interestingly, gene ontology analysis revealed that the genes significantly upregulated during EB differentiation have considerable functions during neural differentiation and development (Table 1) and, as a result, shared >60% of the functions in biological processes with those genes upregulated in TR2/S4+ cells. This becomes more evident when the top 15 functions in the biological process of the gene-ontology list are compared between the two cell-populations as over 70% of them appeared in both lists (Table 1). Similarly, the downregulated genes in both cell populations also shared >60% of gene ontology functions. Therefore, global gene expression analysis does provide certain support for the similar phenotype and differentiation observed between TR2/

S4+ cells and the early EBs (EBs in suspension).

Since cells in the early neural differentiation and early EB formation share a similar gene expression pattern and phenotype, we asked whether early mesendoderm differentiation would also generate this population. To address this question, we differentiated H1 hESCs into definitive endoderm using two different methods (Activin A with sodium butyrate or LY294002, see Materials and Methods for details). Both differentiation procedures exhibited a similar phenotype and gene expression pattern, in which expression of Sox17 and FoxA2 steadily increased over the 3-day differentiation period, whereas, after an initial upregulation, expression of the mesendoderm markers, brachyury and Mixl1, were downregulated by day 3 (Figure 7A). Furthermore, Sox2 expression was continuously downregulated (Figure 7A). This gene expression pattern is consistent with our previous data and existing published data on endoderm differentiation [6,7,27], indicating that the majority of these cells have differentiated to mesendoderm by day 1–2 of the differentiation and are committed to endoderm by day 3. However, during the differentiation, both Tra-1-81 and SSEA4 were continuously expressed in a considerable proportion of cells, approximately 90% and 70% after 2 and 3 days of differentiation, respectively (Figure 7B). These results demonstrated that differentiated mesendoderm and early endoderm cells can also express TRA-1-81 and SSEA4, which is in line with a previous report that found Tra-1-60

and SSEA4 positive cells in a proportion of Sox17+ cells that could only be differentiated into endoderm and mesoderm [28]. As a whole, these results therefore indicate that differentiation towards the neural lineage may share a similar initial process to differentiation via EB formation that is distinct from that of high-dose Activininduced definitive endoderm differentiation.

3 Rapid alloy prototyping: compositional and thermo-mechanical high throughput bulk combinatorial design of structural materials based on the example of 30Mn–1.2C–xAl triplex steels[p]

4.Results

4.1 Mechanical testing of the RAP samples

Following the high throughput procedure outlined above five alloy compositions, each exposed to nine respective heat treatments, were produced, processed and evaluated within 35 h. The mechanical properties of these 45 different material conditions (i.e. in total 135 tensile tests and hardness measurements) are shown as an overview in Fig. 2 in terms of the yield strength (YS) (Fig. 2a), ultimate tensile strength (UTS) (Fig. 2b), total elongation (TE) (Fig. 2c) and hardness (Fig. 2d). The results are plotted according to the systematically varied Al content (Table 1) and colour coded

according to the individual ageing conditions. Pronounced effects associated with the changes in chemical composition and ageing parameters on the mechanical behaviour of the materials can be clearly distinguished. For the reference material (no Al addition, i.e. Fe–30Mn–1.2C) the best mechanical behaviour was observed for the as-homogenised, non-aged state. This alloy was characterised by a YS of 360 MPa, strong work hardening (UTS = 830MPa) and high ductility (TE = 77%). Ageing of the Fe–30Mn–1.2C alloy showed that the YS was virtually unchanged and the hardness slightly increased. However, ageing greatly reduces both the UTS and TE. This embrittlement becomes most apparent for long ageing times (24 h) and higher temperatures (>500°C). For the alloy Fe–30Mn–1.2C–8Al, i.e. the material with the highest Al concentration, the opposite trend applied. Without ageing the mechanical data are similar to those of the Al-free alloy, with only a slight change in YS (increase), UTS and TE (decrease). Ageing treatments for 1 h, however, led to a simultaneous improvement in YS, UTS and hardness (increasing with temperature) and to only a slight drop in TE. Ageing of alloy Fe–30Mn–1.2C–8Al for 24 h further increased YS, UTS and hardness to values almost twice as high as in the ashomogenized state, but also drastically reduced the ductility. The mechanical data for the alloys with intermediate Al contents appear as the superimposition of the two different behaviours described above: as a general rule, the values obtained for the alloys with 2, 4 and 6 wt.%